

Venkat Neelraj Nitta

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SUMMARY

AI/ML engineer with 2+ years building LLM agents, retrieval systems and production ML pipelines in Python.

WORK EXPERIENCE

- **Staples** Framingham, MA, US
AI Engineer Intern Jan 2026 – Jul 2026
 - Maintained and extended the retrieval layer of an agentic e-commerce system: three Databricks Vector Search indexes split by content type and a Neo4j product graph, owning bug fixes and features and adding product-relationship edges.
 - Shipped a fine-tuned multi-label DeBERTa classifier in PyTorch tagging every SKU on three axes, trained on LLM-generated labels with an LLM fallback; about 0.8 accuracy on labeled sets plus an LLM judge.
 - Cut batch inference across roughly 300K products from about 4 hours to 90 minutes by moving it onto Ray for GPU utilization on a single-node cluster, then lowered recurring cost by tagging only changed products.
 - Owned the data path feeding it: SQL consolidating 15+ Snowflake tables into one flattened Delta table, refreshed daily on changed records only (about 5 to 10K of 300K SKUs), with validation in SQL and Python.
- **Cognida.ai** Hyderabad, India
Software/AI Developer Jul 2022 – Aug 2024
 - Built retrieval and embedding-based RAG models for financial document search, evaluating retrieval accuracy and reducing query resolution time from 10 days to minutes.
 - Built unsupervised anomaly detection over millions of health insurance claims (PySpark features, K-means) to surface subrogation cases a rule engine missed; shipped to production, flagging an additional 8% of new cases.
- **Lead Teaching Assistant**, Northeastern University, Boston, MA, US Sep 2026 – Present

SKILLS

Languages & Backend: Python, SQL, Java, FastAPI, Django, REST APIs, Docker, Linux

ML & Deep Learning: PyTorch, Scikit-learn, TensorFlow, Hugging Face, Quantization, Ray

LLMs, Agents & Retrieval: LangGraph, LangChain, RAG, Vector Search, Embeddings, FAISS

Data & Cloud: Databricks, Delta Lake, Snowflake, PySpark, Neo4j, AWS, Azure, CI/CD

Engineering Practice: Git, AI-Assisted Development (Claude Code, Cursor), Unit & Integration Testing

PROJECTS

- **Automobile Insurance Assistance (Ollama, YOLOv11, Django) [GitHub]:** Local-first RAG over insurance policies plus vehicle-damage segmentation, behind a pluggable local LLM interface with token streaming.
- **Artha AI Portfolio Allocation (CNN, gunicorn, DVC) [GitHub]:** Model that emits portfolio weights through a softmax, trained on a custom Sharpe-based objective and served as a REST API from one multi-stage container image.

EDUCATION

- **Northeastern University** Boston, MA, US
Master of Professional Studies in Applied Machine Intelligence (STEM); GPA: 3.92 Sep 2024 – Dec 2026 (Expected)
 - Relevant Coursework: Fundamentals of AI, AI System Technologies, Data Management & Big Data
- **Mahindra Ecole Centrale** Hyderabad, India
Bachelor of Technology in Electrical and Electronics Engineering Aug 2018 – Jun 2022
 - Relevant Coursework: Machine Learning, Data Structures, Big Data Computing

AWARDS & CERTIFICATIONS

- **Hackathon:** Won the Workhuman-sponsored hackathon (Lighthouse) [GitHub]. **Certifications:** Generative AI with Large Language Models (DeepLearning.AI); AI Agents on Databricks